Free Game Tracker

CS 193 — Homework 9: Final Project Design Document

1.1 — Problem Statement

As someone who regularly plays video games, I find it frustrating to keep track of limited-time free game offers across different gaming platforms. Checking websites like Epic Games Store and other giveaway sources manually takes time, and it is easy to miss offers before they expire. I want to solve this problem because I enjoy collecting games and would benefit from having one place to keep track of available free games.

1.2 — Proposed Solution

I plan to build a Free Game Tracker application that collects and displays information about limited-time free video game offers from platforms such as Epic Games Store and Steam. The application will show available games and their offer expiration dates in one place, making it easier for users to find and claim free games before the offers end.

2.1 — Eisenhower Matrix and MVP

The Eisenhower Matrix helps prioritize which features should be included in the first version of the Free Game Tracker and which can be added in future versions.

	Urgent	**Not Urgent**
Important	DO — MVP 1. Fetch current free game giveaways from an online API. 2. Display game titles, platforms (Steam/Epic), and expiration dates. 3. Show users a link to claim each available game.	SCHEDULE — V2.0+ 1. Send notifications when new free games become available. 2. Allow users to mark games as claimed. 3. Save a history of previously available giveaways.
Not Important	DELEGATE — Use a Library 1. Use an HTTP client library to handle internet requests.	DELETE — Cut It 1. Integrate directly with Steam or Epic accounts.

	2. Use a JSON parsing library to read game data. 3. Use built-in date/time utilities to handle expiration dates.	2. Automatically claim free games for users. 3. Create user profiles, leaderboards, or social features.

2.2 — Tech Stack

- **Programming Language:** Java
- **Development Environment:** IntelliJ IDEA
- **Libraries:** Java HttpClient for API requests, Gson for JSON parsing, and Java date/time utilities for handling expiration dates (proposed).
- **Data Storage:** JSON or text files for simple local storage if needed.
- **External Data Source:** GamerPower API to retrieve information about free game giveaways from Steam and Epic Games Store. This is a tentative choice pending testing.
- **Version Control:** Git and GitHub for tracking changes, organizing files, and documenting development.

2.3 — Chunked Build Plan

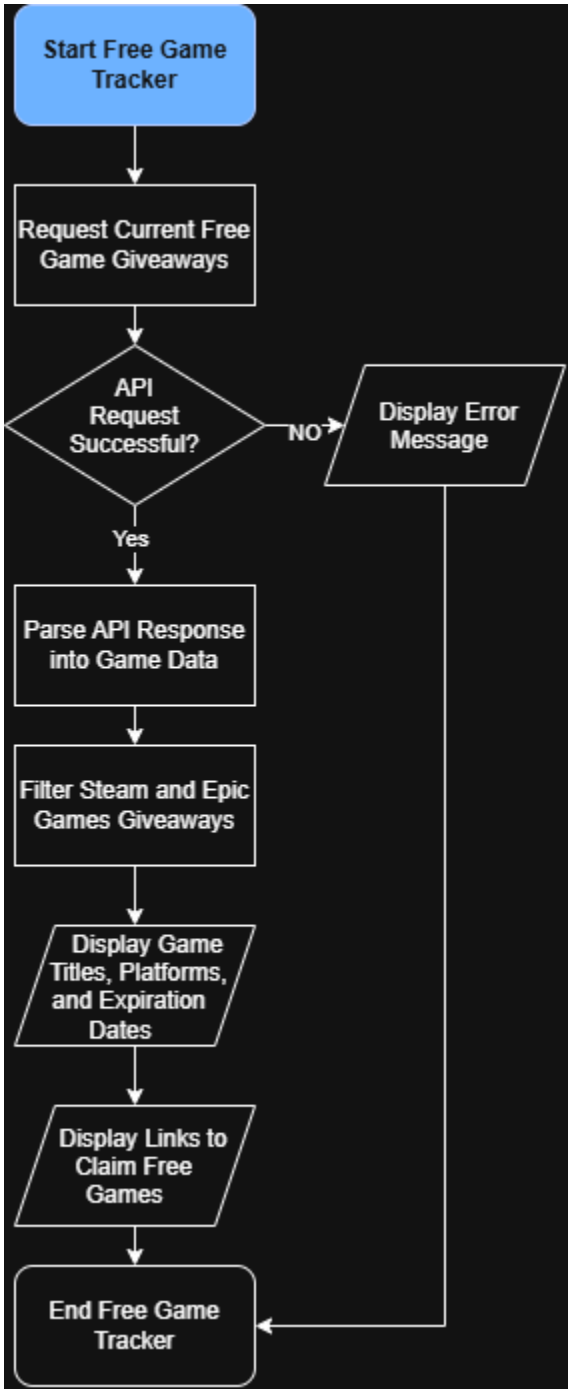
Each milestone represents approximately one meaningful Git commit.

1. Set up the Java project in IntelliJ IDEA and initialize a GitHub repository with a README.
2. Create the basic Java application structure and a main menu for the Free Game Tracker.
3. Create a Game class to store information such as game title, platform, expiration date, and claim link.
4. Test a free game giveaway API and implement a method to retrieve current giveaway data.
5. Parse the retrieved data and convert it into Game objects that the application can use.
6. Display the available free games in a readable format, including their titles, platforms, and expiration dates.
7. Add the ability to view a game's claim link and handle errors such as failed API requests or missing information.
8. Test the completed application, fix bugs, and update the README with setup instructions and documentation.

2.4 — System Diagram

The system diagram illustrates the planned workflow of the Free Game Tracker application. The program requests giveaway information from an external source, checks whether the request succeeds, processes the returned data, and displays available Steam and Epic Games giveaways along with their expiration dates and claim links.

If the request fails, the application displays an error message.



3.1 — GitHub Repository and README

The Free Game Tracker project is hosted in a public GitHub repository. The repository contains a README describing the project goal, MVP scope, technology stack, and future development plans.

The repository also includes folders for source code, testing, and project documentation. Git commits will be used to track progress throughout development.

GitHub Repository:

https://github.com/singadi26/FreeGameTracker

Project Status

The project is currently in the planning and design phase. Development will begin with the core MVP features, with additional functionality considered after the basic application is working.